Gated Data-Space Inversion for Reservoir Prediction

Reducing variance loss in posteriors

Masters dissertation ▷ Thiago Tomás de Paula

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Introduction

Theory framework

Prediction forward model

The prediction forward model, or forward model, is the underlying mapping between a model's parameters, controls and initial states to its observable features.

Denoting by $\mathbf{m}$ the vector of model parameters, and by $\mathbf{u}$ the set of controls applied on the field, and by $\mathbf{d}$ the reservoir production within the time frame of the controls, one can write

$$\mathbf{d} = \mathbf{g}(\mathbf{m}, \mathbf{u}; \mathbf{x}_0) \tag{1}$$

with $\mathbf{x}_0$ the initial value of reservoir states (usually omitted).

Data features

In this work, the features inside $\mathbf{d}$ are the following well measurements:

▷ oil and water production rates	(OPR and WPR)	[m^3/day]
▷ water injection rate	(WIR)	[m^3/day]
▷ bottomhole pressure	(BHP)	[Bar]
▷ water cut percentage	(WCTP)	[%]

Field oil production total

Field oil production total (FOPT, SI unit of m^3) is also relevant to this work, despite not being part of $\mathbf{d}$.

The key assumption behind the forward model is that it is deterministic. Implementation wise, g consists of evaluating a set of equations at each gridblock of the model.

Each evaluation evolves the model a little further in time, until the scheduled time period is appreciated, concluding the process.

Black-oil equations are the most common implementations of the forward model [2] due to their fair simplicity and accuracy. The fluid is assumed composed of three liquid phases, water, oil and gas.

A given phase α's component mass A_α satisfies the conservation equation

$$\frac{\partial}{\partial t}\left(\phi_{\text{ref}} A_\alpha\right) + \nabla \cdot \mathbf{u}_\alpha + q_\alpha = 0, \tag{2}$$

where

$$\begin{aligned}
A_w &= m_\phi b_w s_w, & \mathbf{u}_w &= b_w \mathbf{v}_w, & \text{(3a)}\\
A_o &= m_\phi (b_o s_o + r_{og} b_g s_g), & \mathbf{u}_o &= b_o \mathbf{v}_o + r_{og} b_g \mathbf{v}_g, & \text{(3b)}\\
A_g &= m_\phi (b_g s_g + r_{go} b_o s_o), & \mathbf{u}_g &= b_g \mathbf{v}_g + r_{go} b_o \mathbf{v}_o. & \text{(3c)}
\end{aligned}$$

and the phase fluxes are given by Darcy's law:

$$\mathbf{v}_\alpha = -\lambda_\alpha \mathbf{K}(\nabla p_\alpha - \rho_\alpha \mathbf{g}). \tag{4}$$

Finally, the following closure equations are imposed:

$$s_w + s_o + s_g = 1, \qquad p_{c,ow} = p_o - p_w, \qquad p_{c,og} = p_o - p_g. \tag{5}$$

Predicting versus forecasting

Prediction refers to the estimation of any datum at a given time, while forecasting refers to the estimation of any datum beyond the observation period. Forecasting is a type of prediction, but in particular prediction of new data.

History and validation

In Data Science literature, a common way of estimating forecast quality is splitting available data into a training partition and a test partition. The training partition is used to freely adjust predictions within a given period, and the test, to verify how close predictions will be on new data.

In the context of this work, data from a given reference will be split into two disjointed periods in time: **history** period, and **validation** period, mirroring the train and test splits.

Let $\mathbf{d}$ denote the true production data vector of a reference reservoir across time, free of observation error. Data is separated into a history period and a validation period, so that

$$\mathbf{d} = \begin{bmatrix} \mathbf{d}_h^\top & \mathbf{d}_v^\top \end{bmatrix}^\top \tag{6}$$

with $\mathbf{d}_h$ the ground-truth history data, and $\mathbf{d}_v$ the ground-truth validation data.

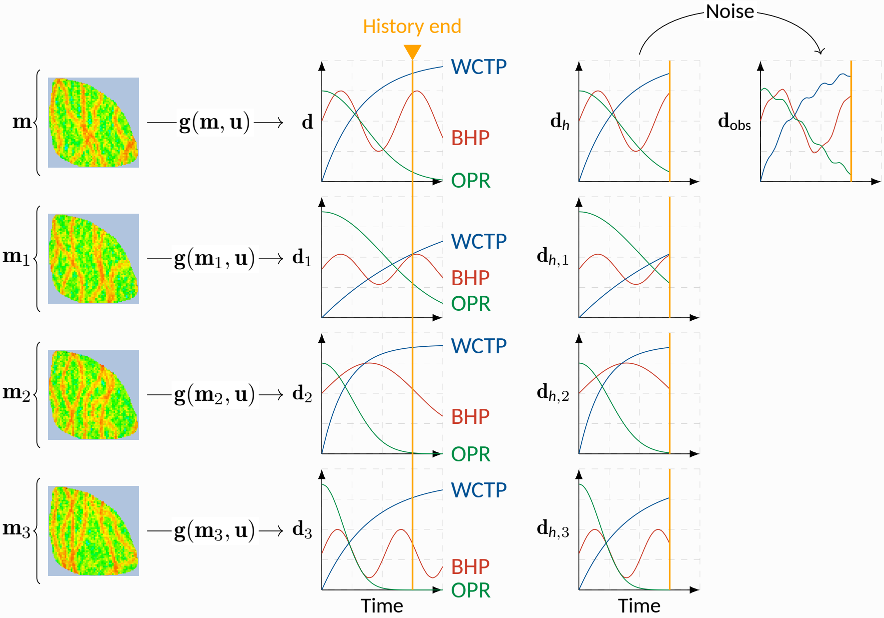
Let $\mathbf{u}$ and $\mathbf{m}$ be the controls applied to the ground-truth and the parameters of the ground-truth, respectively. Then

$$\mathbf{d} = \mathbf{g}(\mathbf{m}, \mathbf{u}). \tag{7}$$

Observed data is $\mathbf{d}_h$ corrupted by measurement noise:

$$\mathbf{d}_{\mathrm{obs}} = \mathbf{d}_h + \mathcal{N}(\mathbf{0}, \mathbf{C}_e), \tag{8}$$

where $\mathbf{C}_e$ is the noise covariance matrix.


History end
Noise
$\mathbf{m}$
$\mathbf{g}(\mathbf{m}, \mathbf{u})$
$\mathbf{d}$
WCTP
BHP
OPR
$\mathbf{d}_h$
$\mathbf{d}_{\text{obs}}$
$\mathbf{m}_1$
$\mathbf{g}(\mathbf{m}_1, \mathbf{u})$
$\mathbf{d}_1$
$\mathbf{d}_{h,1}$
$\mathbf{m}_2$
$\mathbf{g}(\mathbf{m}_2, \mathbf{u})$
$\mathbf{d}_2$
$\mathbf{d}_{h,2}$
$\mathbf{m}_3$
$\mathbf{g}(\mathbf{m}_3, \mathbf{u})$
$\mathbf{d}_3$
$\mathbf{d}_{h,3}$
Time
Time

DSI-ESMDA can be conceptulized as a Bayesian approach to the forecasting problem, aiming to estimate the underlying conditional probability distribution of predictions given the reference data.

The DSI-ESMDA problem

Find the posterior distribution

$$p(\mathbf{d}^* \mid \mathbf{d}_{\text{obs}}) \propto \mathcal{L}(\mathbf{d}_{\text{obs}} \mid \mathbf{d}^*) p(\mathbf{d}^*). \tag{9}$$

where $\mathbf{d}^*$ maximizes $p(\mathbf{d}^* \mid \mathbf{d}_{\text{obs}})$.

For given data, the algorithm returns samples from this posterior distribution, outputing a range of values rather than a single value. While this makes evaluating the algorithm less straightforward, it allows for quantification of uncertainty.

Ensemble Smoothers share this approach at a conceptual level, but they perform a single maximization. This leads to poor matches to real data if assumptions on the likelihood $\mathcal{L}$ are not met.

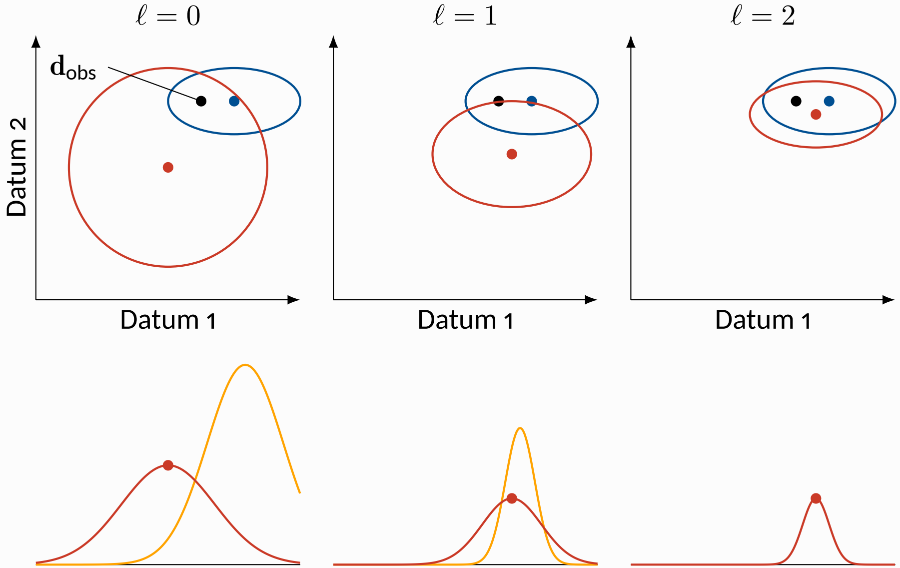
Multiple Data Assimilation mitigates this problem by making N_a optimizations with resampled reference data:

$$p^1(\mathbf{d}^* \mid \mathbf{d}^1_{\text{obs}}) \propto \mathcal{L}(\mathbf{d}^1_{\text{obs}} \mid \mathbf{d}^*)p(\mathbf{d}^*) \tag{10}$$

$$p^2(\mathbf{d}^* \mid \mathbf{d}^2_{\text{obs}}, \mathbf{d}^1_{\text{obs}}) \propto \mathcal{L}(\mathbf{d}^2_{\text{obs}} \mid \mathbf{d}^*)p^1(\mathbf{d}^* \mid \mathbf{d}^1_{\text{obs}}) \tag{11}$$

$$\vdots \tag{12}$$

$$p^{N_a}(\mathbf{d}^* \mid \mathbf{d}^{N_a}_{\text{obs}}, \ldots, \mathbf{d}^1_{\text{obs}}) \propto \mathcal{L}(\mathbf{d}^{N_a}_{\text{obs}} \mid \mathbf{d}^*)p^{N_a-1}(\mathbf{d}^* \mid \mathbf{d}^{N_a-1}_{\text{obs}}, \ldots, \mathbf{d}^1_{\text{obs}}) \tag{13}$$


$\ell = 0$
$\ell = 1$
$\ell = 2$
$\mathbf{d}_{\text{obs}}$
Datum 2
Datum 1
Datum 1
Datum 1

$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^{\ell} + \mathbf{K}^{\ell}(\mathbf{d}_{\text{obs}} + \sqrt{\alpha^{\ell}}\mathbf{e}_j^{\ell} - \mathbf{d}_{h,j}^{\ell}), \tag{14a}$$

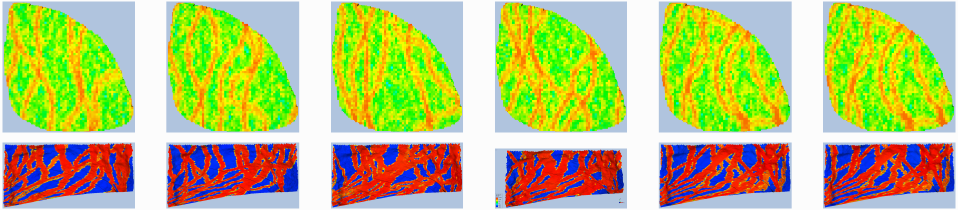
$$\mathbf{K}^{\ell} = \Delta\mathbf{D}^{\ell}\Delta\mathbf{D}_h^{\ell}(\Delta\mathbf{D}_h^{\ell}\Delta\mathbf{D}_h^{\ell} + \alpha^{\ell}\mathbf{C}_e)^{-1}. \tag{14b}$$

$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^{\ell} + (\mathbf{R} \circ \mathbf{K}^{\ell})(\mathbf{d}_{\text{obs}} + \sqrt{\alpha^{\ell}}\mathbf{e}_j^{\ell} - \mathbf{d}_{h,j}^{\ell}), \tag{15}$$

$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^{\ell} + (\mathbf{R} \circ \mathbf{K}^{\ell})[\boldsymbol{\alpha}_j^{\ell} \circ (\mathbf{d}_{\mathsf{obs}} - \mathbf{d}_{h,j}^{\ell})], \tag{16a}$$

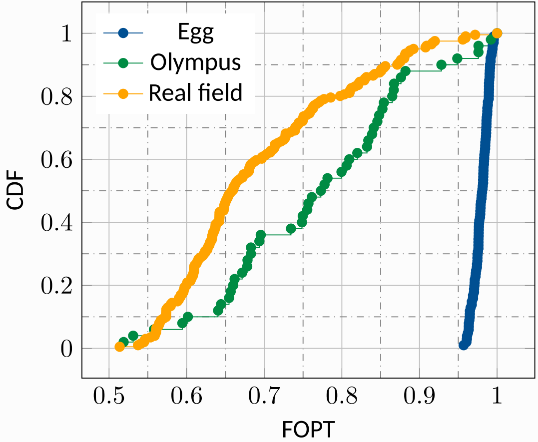
$$\alpha_{j,v}^{\ell} = \operatorname{erf}\left(\frac{1}{\lambda\sqrt{2\sigma_v^2}}|(\mathbf{d}_{\mathsf{obs}} - \mathbf{d}_{h,j}^{\ell})_v|\right) \tag{16b}$$

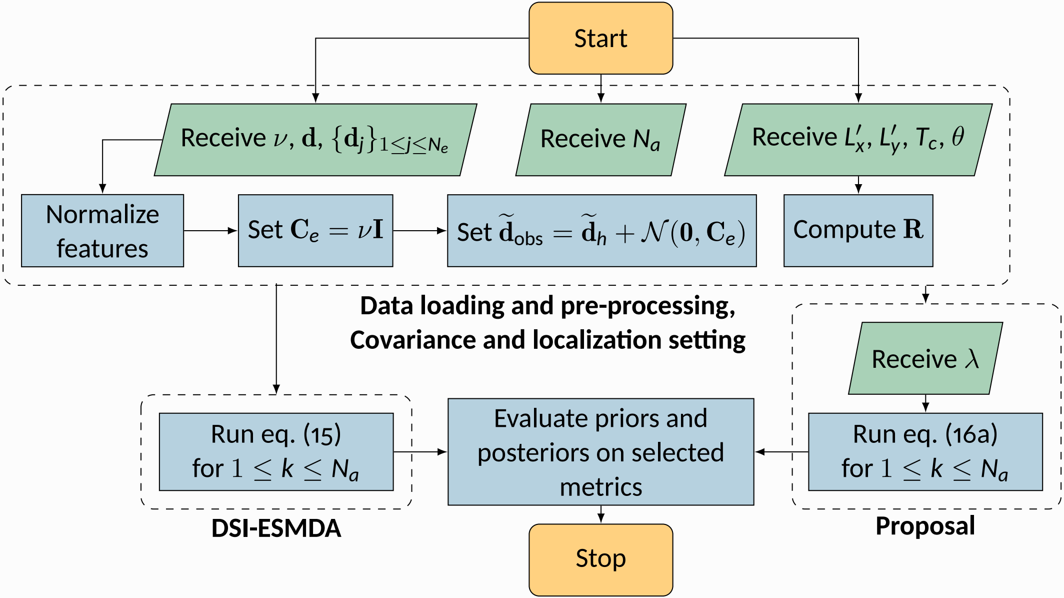
Methods



Property	Egg	Olympus	Real field	Unit
Rock compressibility	0.0	1.42×10^{-10}	-	1/Pa
Water compressibility	1.0×10^{-10}	3.97×10^{-10}	-	1/Pa
Oil dynamic viscosity	5.0×10^{-3}	$2.8\text{-}3.5 \times 10^{-3}$	-	Pa·s
Water dynamic viscosity	1.0×10^{-3}	0.398×10^{-3}	-	Pa·s
Well diameter	20	19	-	cm
Wells	8(P) 4(I)	11(P) 7(I)	16(P) 8(I)	-
Active cells	18,533	192,750	-	-
Cell size	$8\times8\times4$	$50\times50\times3$	-	$m\times m\times m$
Ensemble size	101	50	201	-

The field oil production total (FOPT) is a measurement commonly used to compare ensembles since ground-truth and realizations are each represented by a single value: how much oil was produced within the period of interest.




Start
Receive ν, $\mathbf{d}$, $\{\mathbf{d}_j\}_{1\leq j\leq N_e}$
Receive N_a
Receive L'_x, L'_y, T_c, θ
Normalize features
Set $\mathbf{C}_e = \nu\mathbf{I}$
Set $\widetilde{\mathbf{d}}_{\text{obs}} = \widetilde{\mathbf{d}}_h + \mathcal{N}(\mathbf{0}, \mathbf{C}_e)$
Compute $\mathbf{R}$
Data loading and pre-processing, Covariance and localization setting
Receive λ
Run eq. (15) for $1 \leq k \leq N_a$
Evaluate priors and posteriors on selected metrics
Run eq. (16a) for $1 \leq k \leq N_a$
DSI-ESMDA
Proposal
Stop

Four metrics are considered on the evaluation of posterior/prior distributions [1, subsection 9.4.2]: history observation coverage, validation observation coverage, mean squared error, and data-mismatch.

Denoting an ensemble by $\mathbf{Y} = \begin{bmatrix} \mathbf{y}_1 & \cdots & \mathbf{y}_{N_e} \end{bmatrix}$, these metrics are defined as:

$$\mathrm{OC}_h(\mathbf{Y}) = \frac{1}{N_h} \sum_{i=1}^{N_h} \mathbb{1}\left(\min(\mathbf{Y})_i \le d_{h,i} \le \max(\mathbf{Y})_i\right) \tag{17}$$

$$\mathrm{OC}_v(\mathbf{Y}) = \frac{1}{N_v} \sum_{i=1}^{N_v} \mathbb{1}\left(\min(\mathbf{Y})_i \le d_{v,i} \le \max(\mathbf{Y})_i\right) \tag{18}$$

$$\mathrm{MSE}(\mathbf{Y}) = \frac{1}{N_h}(\mathbf{d}_h - \overline{\mathbf{y}})^\top \mathbf{C}^{-1}(\mathbf{d}_h - \overline{\mathbf{y}}), \quad \mathbf{C} = \Delta\mathbf{Y}\Delta\mathbf{Y}^\top + \mathbf{C}_e \tag{19}$$

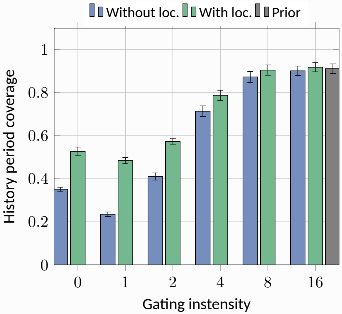
$$\mathcal{O}(\mathbf{y}_j) = \frac{1}{2N_h}(\mathbf{d}_h - \mathbf{y}_{j,h})^\top \mathbf{C}_e^{-1}(\mathbf{d}_h - \mathbf{y}_{j,h}) \tag{20}$$

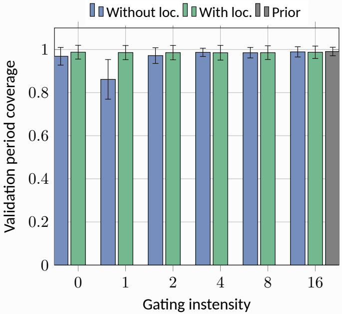
Reservoir	N_e	Number of time steps	Time step size	Simulation life		History split
Egg	100	120	30 days	3600 days	(9.8 years)	60%
Olympus	49	20	1 calendar year	7305 days	(20 years)	60%
Real field	200	510	1 calendar month	12785 days	(35 years)	60%*

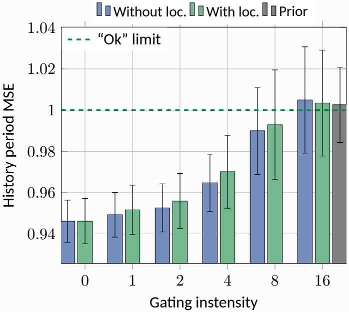
Reservoir	Critical length in x	Critical length in y	Critical time	Anisotropy
Egg	100	100	3000 days	0°
Olympus	100	100	7200 days	0°
Real field	-	-	10500 days	0°

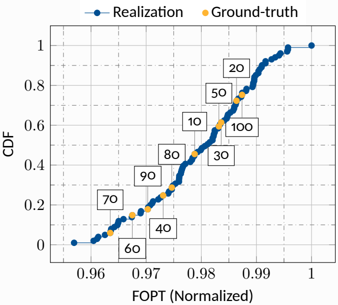
Reservoir	Well features used		Number of wells		Total features
	Producer	Injector	Producer	Injector	
Egg	OPR, WPR	BHP	4	8	16
Olympus	OPR, WPR	WIR	11	7	29
Real field	BHP, WCTP	BHP	16	8	39 (a WCTP was constant)

Results

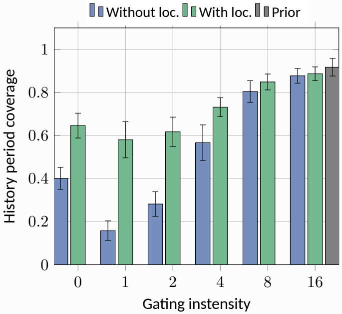

Without loc.
With loc.
Prior
History period coverage
1
0.8
0.6
0.4
0.2
0
0
1
2
4
8
16
Gating instensity

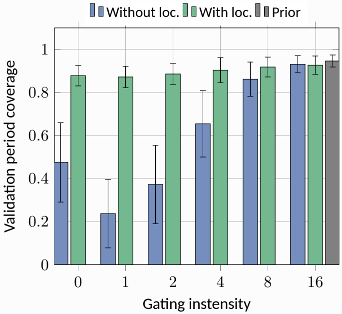

Without loc.
With loc.
Prior
Validation period coverage
1
0.8
0.6
0.4
0.2
0
0
1
2
4
8
16
Gating instensity

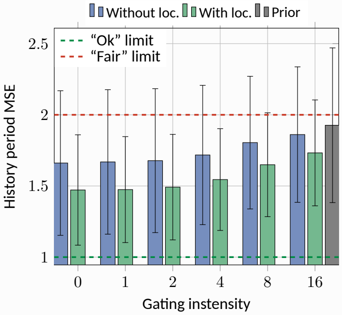

Without loc.
With loc.
Prior
"Ok" limit
History period MSE
1.04
1.02
1
0.98
0.96
0.94
0
1
2
4
8
16
Gating instensity

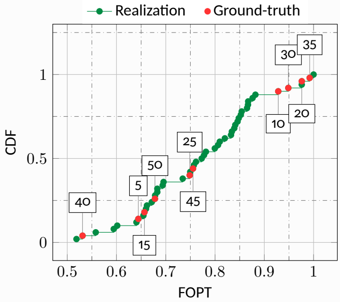

Realization
Ground-truth
CDF
1
0.8
0.6
0.4
0.2
0
20
50
10
100
80
30
90
70
40
60
0.96
0.97
0.98
0.99
1
FOPT (Normalized)

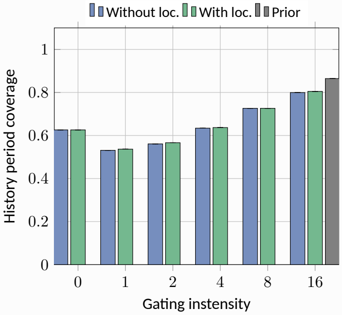
Results ▶ Olympus case

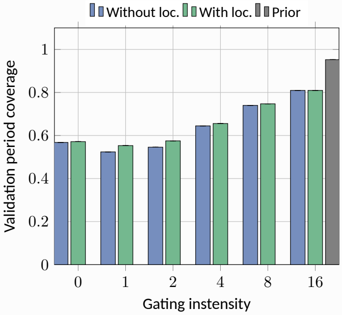


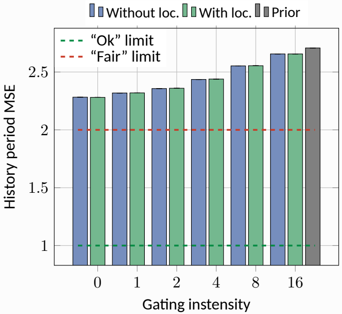


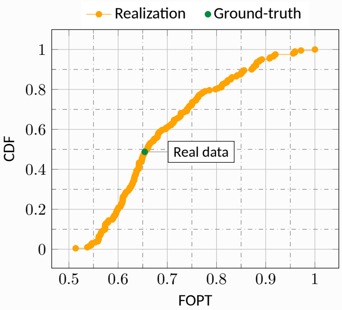





Without loc.
With loc.
Prior
History period coverage
1
0.8
0.6
0.4
0.2
0
0
1
2
4
8
16
Gating instensity


Without loc.
With loc.
Prior
Validation period coverage
1
0.8
0.6
0.4
0.2
0
0
1
2
4
8
16
Gating instensity


Without loc.
With loc.
Prior
"Ok" limit
"Fair" limit
History period MSE
2.5
2
1.5
1
0
1
2
4
8
16
Gating instensity


Realization
Ground-truth
Real data
CDF
1
0.8
0.6
0.4
0.2
0
0.5
0.6
0.7
0.8
0.9
1
FOPT

Discussion

OC, MSE and mismatch increase

Increase in gating intensity tended to increase observation coverage, mean squared error, and data mismatch, albeit with case-dependent sensitivities.

OC, MSE and mismatch increase

Localization yielded mixed results: for the Egg and real field cases, effects on coverage and MSE were minimal, especially for intense enough gatings, while the effect in the Olympus case was drastic. Cumulative distribution plots of the field oil production total suggest that ground-truths are unbiased in the Egg and real field cases, but somewhat biased for the Olympus case.

Mild and small gating intensities versus standard DSI-ESMDA

Mild gating intensities (2 in the Egg and Olympus cases, 4 in the real field case) produced similar results to the standard DSI-ESMDA for same localization configuration, while small gating (1 for the Egg case, 1 and 2 for the Olympus and real field cases) had worse coverages than the standard DSI-ESMDA.

Conclusion

Bibliography I

[1] Alexandre Anozé Emerick. *Ensemble Data Assimilation Applied to Geological Reservoir Models*. 1st ed. Rio de Janeiro, RJ, Brazil: Universidade Petrobras, June 2025. ISBN: 978-65-88763-29-2.

[2] Atgeirr Flø Rasmussen et al. "The Open Porous Media Flow reservoir simulator". In: *Computers & Mathematics with Applications* 81 (2021). Development and Application of Open-source Software for Problems with Numerical PDEs, pp. 159–185. ISSN: 0898-1221. DOI: `https://doi.org/10.1016/j.camwa.2020.05.014`.

Gated DSI-ESMDA

Reducing variance loss in posteriors

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August 4th, 2026

"It's hard to make predictions, specially about the future"

— Yogi Berra

Thank you!